

Newton MPM Performance Benchmark Report

Executive Summary

This report presents the performance analysis of Newton's Material Point Method (MPM) particle simulation system. The benchmark was conducted on 2025-08-27 10:41:14 testing 2 different configurations. **Key Findings:**

- Best throughput: 10,065,991 particles/second
- Average GPU utilization: 0.0%
- Memory usage range: 0.0 - 0.0 GB
- Configurations tested: 2

System Configuration

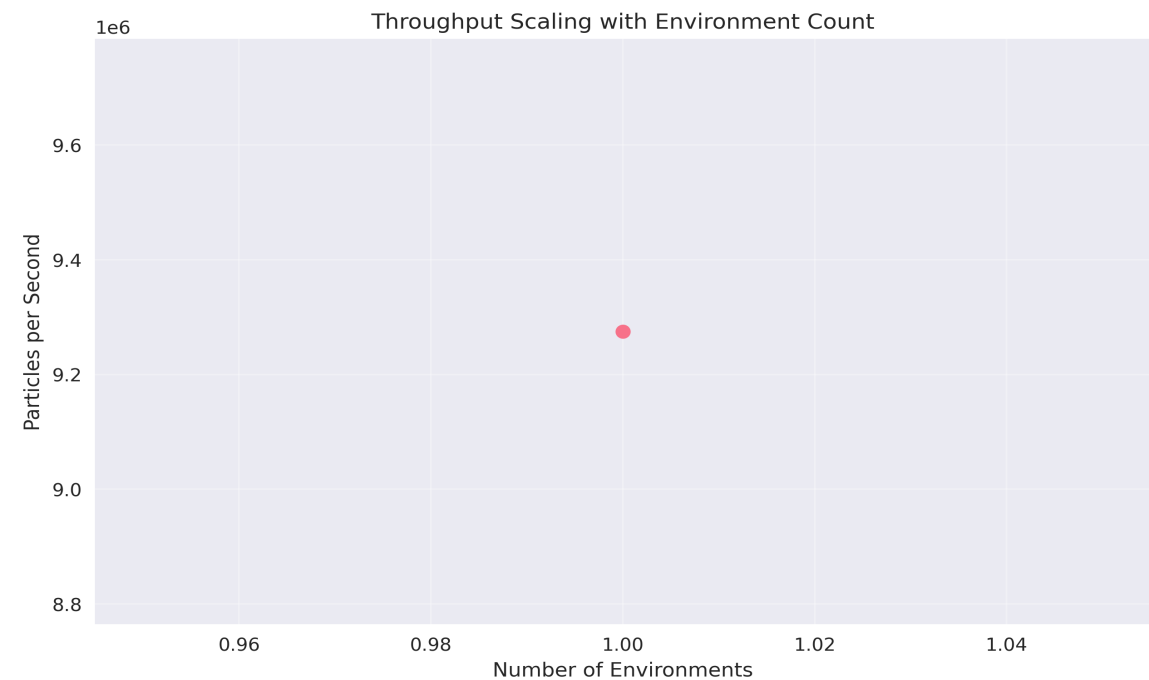
Hardware:

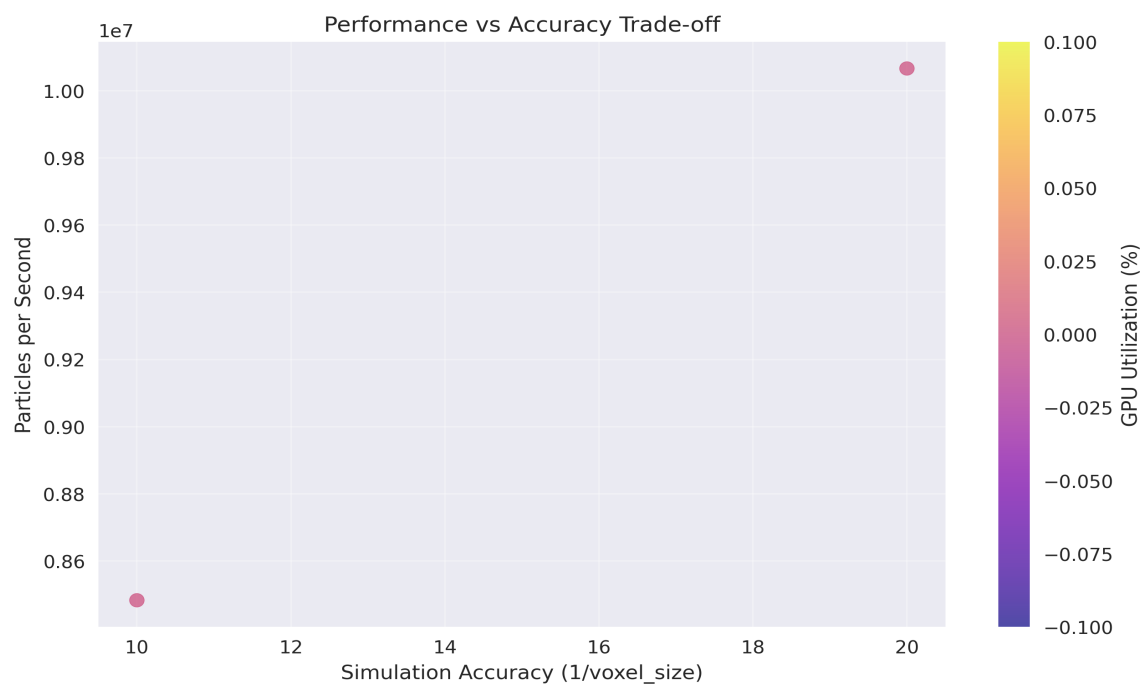
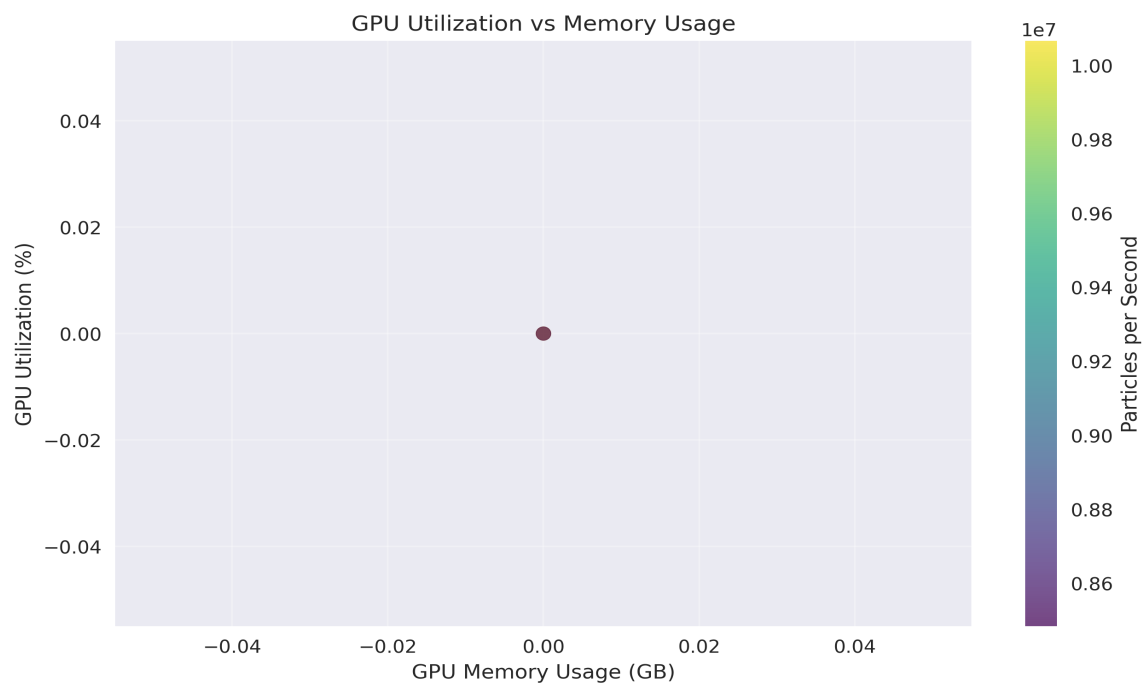
- CPU: AMD Ryzen 9 5900X 12-Core Processor (12 cores)
- Memory: 31.3 GB
- GPU: cuda:0

Software:

- Warp Version: Unknown
- Newton Version: 0.1.1.a2
- GPU Monitoring: Unavailable

Performance Analysis





Optimization Recommendations

1. Best Throughput Configuration: voxel_size=0.05, particles_per_cell=3, achieving 10,065,991 particles/second
2. Best Efficiency Configuration: voxel_size=0.05, particles_per_cell=3, efficiency score: 0
3. Low GPU utilization detected. Consider increasing particle count, reducing voxel size, or adding more environments to better utilize GPU resources.

Detailed Results

Configuration	Particles/sec	GPU Util %	Memory GB
voxel=0.05, envs=1	10,065,991	0.0	0.00
voxel=0.1, envs=1	8,483,277	0.0	0.00